Six Months with AMD Strix Halo

Local AI, Open Source, and Hardware Reality

A Six Month Review of a Strix Halo Rig

linux | ROCm | Local AI | Home Assistant | Coding

winmutt | June 2026
github.com/winmutt

What We're Covering

■ The impulse purchase that changed everything
■■ Hardware reality check (spoiler: it's complicated)
■ Linux + ROCm = Pain (but beautiful pain)
■ Performance: How fast can my AI think?
■ I cut the Alexa cord (and replaced it with OSS)
■ Vibe-coded an entire workspace system
■ From digital to physical: Concrete signs?
■ How buying hardware got me back in open source
■ What I'd do differently (and what I'd do again)

Power Reality: Corsair 300 vs Alternatives

Max Energy Usage Comparison

Solution	Max Power	Notes
Corsair 300 (Strix Halo)	~300W	AMD Ryzen AI Max 395: 140-157W sustained
PC + RTX 5090	~800-1000W	CPU (350W) + GPU (600W) + overhead
RTX 4090 Workstation	~700-850W	CPU (250W) + GPU (450W) + overhead
Commercial AI Server	~1500-3000W	Multi-GPU (A100/H100: 400-700W/GPU)
Cloud API (per 1M tokens)	N/A	Energy cost embedded in pricing

The Efficiency Win

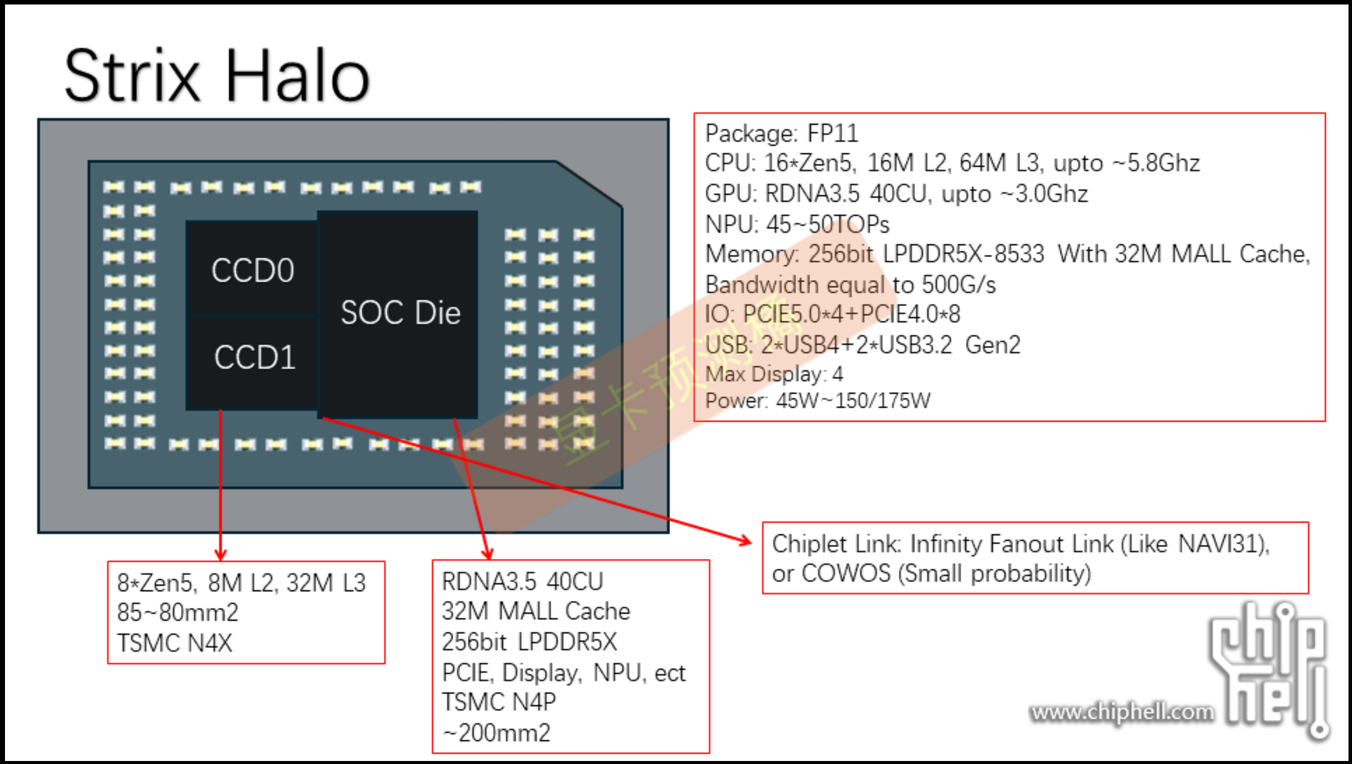
- Strix Halo: ~3x more efficient than RTX 5090 setup
- Single APU vs discrete CPU+GPU power domains
- No datacenter electricity bill
- 128GB unified memory at desktop power envelope

Sources:

- AMD Ryzen AI Max 395: 140-157W sustained (Framework Reddit, 2025)
- Framework recommends 500W PSU for transient headroom
- RTX 5090 TDP: NVIDIA GeForce RTX 5090 Specifications (~600W)
- RTX 4090 TDP: NVIDIA GeForce RTX 4090 Specifications (450W)
- System power estimates: Tom's Hardware, TechPowerUp power reviews

The Hardware: AMD Ryzen AI Max+ 395

APU Die Architecture



Physical Layout

- 16 cores (12 performance + 4 efficiency)
- 2x Core Complex Dies (CCD) with 64MB L3 cache each (community analysis)
- 768KB L1d + 512KB L1i + 16MB L2
- Integrated RDNA 3.5 GPU (40 CUs)
- 128GB LPDDR5X-8000 unified memory (soldered)
- Source: AMD Ryzen AI Max+ 395 teardown analysis (Chiphell, TechPowerUp)

Feat #1070: NUMA Tools Don't See Dual CCD

Feature Request
Traditional NUMA tools detect only 1 node. numactl core pinning doesn't work. Need custom core affinity for dual CCD.

Hardware Reality

• One processor, but 2x core dies with separate 64MB L3 caches

• Traditional NUMA tools detect only 1 node (not 2 CCDs)

• numactl, NUMATopologyFilter can't see CCD boundaries

• llama-server threads not pinned to specific cores

The Solution (Proposed)

• Manual core pinning (like Red Hat's vcpu_pin_set approach)

• Reserve cores per CCD, pin llama-server accordingly

• Pin based on:

- Number of models loaded

- CCD boundaries (manual NUMA mapping)

• Prevent cache-crossing penalties

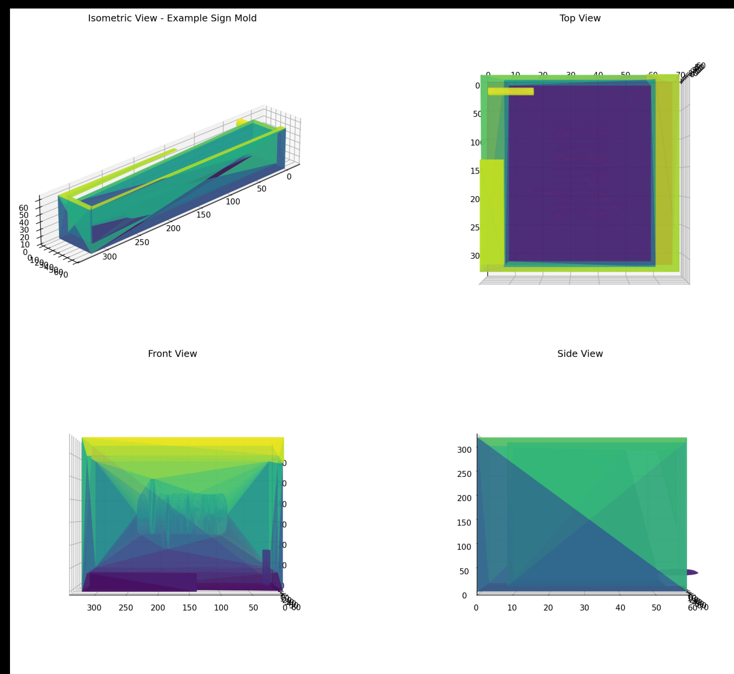
GitHub Feature Request
github.com/lemonade-sdk/lemonade/issues/1070
Status
Open (custom NUMA mapping needed)

Blender: Concrete Sign Mold

December 1, 2025

> *"I am going to use it to create a 3d printed cast for a silicone mold so I can some desktop signs out of concrete."*

Concrete Sign Mold Design



The Process

1. AI generates OpenSCAD design
2. Export to STL
3. 3D print mold

4. Create silicone mold
5. Cast concrete

Cutting the Alexa Cord

LineageOS on Echo: Because I Said So

December 2025
Started exploring XDA Forums for Echo device hacking

The Process:

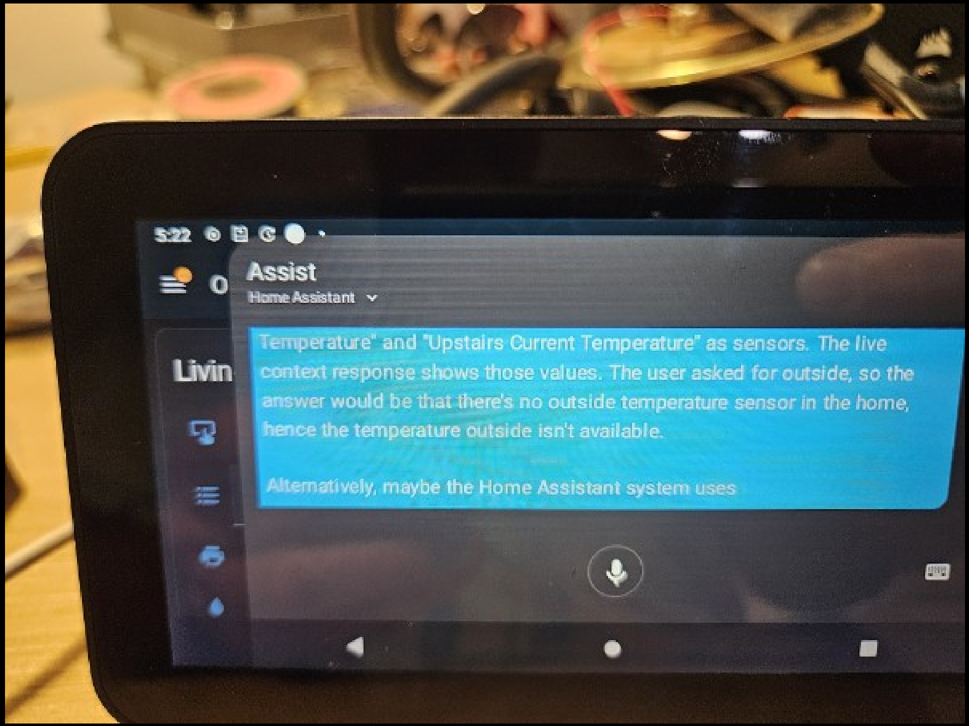
1. Unlock bootloader via amonet exploit
github.com/R0rt1z2/amonet
2. Install TWRP recovery
3. Flash LineageOS 18.1 (Android 11)
4. Configure ViewAssist for Home Assistant
dinki.github.io/View-Assist

> ***Philosophy***
> *"Basically anything with a USB port. If there is a port, there is a way."*

Alexa Replacement Timeline

Dec 6	Started exploring XDA forums
Dec 7	HA running on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything works (except wake word)
Jan 25	Hey Jarvis to the rescue
Jan 29	End-to-end complete
Jan 7, 2026	Issue #4: Echo 8 mic dies after 24h
Feb 2026	Still debugging (it's fine)

Home Assistant on Echo Show



Issue #4: The Echo 8 Mic That Quit

Audio stops working after ~24 hours

Filed
January 7, 2026
Repository
https://github.com/amazon-oss/releases/issues/4
Status
Open (still waiting, it's fine)

The Problem

• Echo 5: Stable for days

• Echo 8: Dies after ~24 hours

• *Root cause: Unknown (yet)*

TL;DR
I participate in open source bugs now. You're welcome.

Project WWS: I Vibe-Coded an Entire System

Winmutt Work Spaces

github.com/winmutt/wws

> ***May 12, 2026***
> *"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."*

What is WWS?

• Remote workspace provisioning

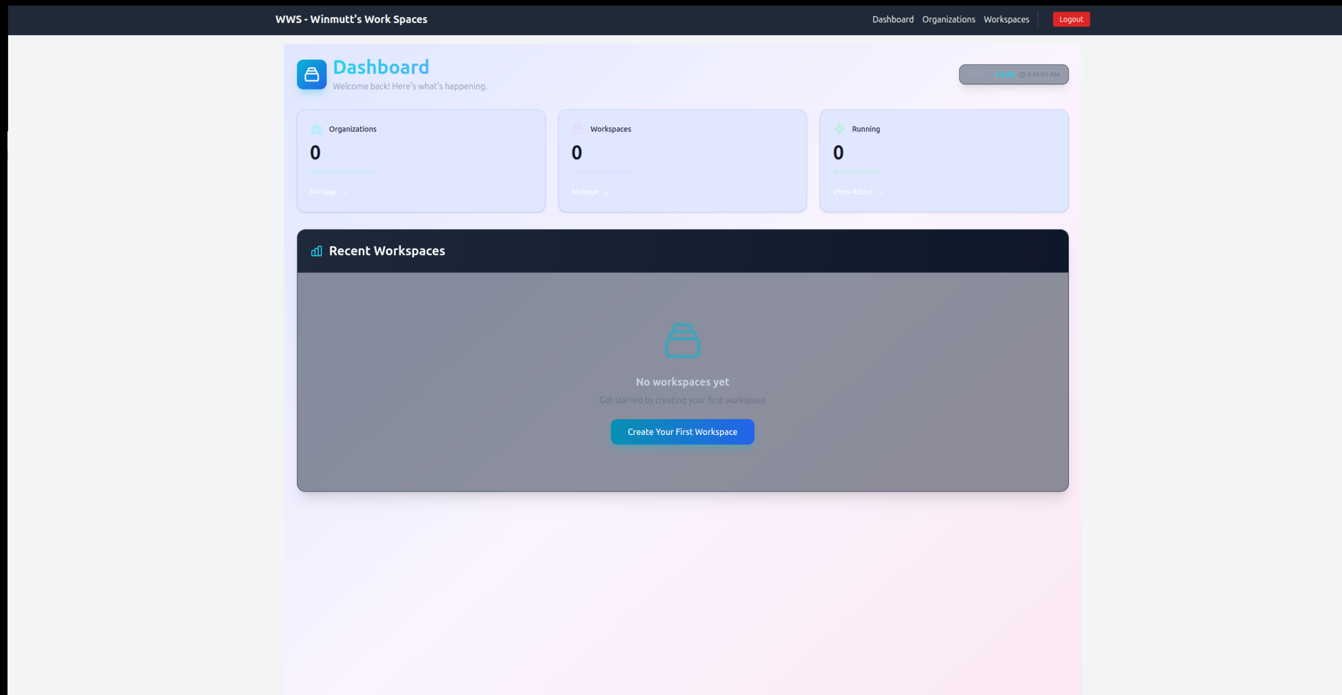
• KVM/Podman isolated environments

• code-server (VSCode in browser)

• GitHub OAuth + RBAC

• First commit: Feb 22, 2026

WWS Dashboard



Lessons Learned (Mostly)

What Worked ✓

- Cline + Qwen3-Coder: Actually produces code
- Prompt Caching: Things hum now
- AMD GPU Libraries: 2x faster (worth the pain)
- WWS Project: Vibe-coded from Feb 22 (first commit) to May 12
- Home Assistant: Alexa is dead, long live OSS

What Didn't ✗

- Memory: 32GB reserved for Lemonade/llama.cpp/opencode
- Ollama: Still crashes after updates
- NPU: Coming soon™ (always)
- Context >64k: TPS drops like a stone

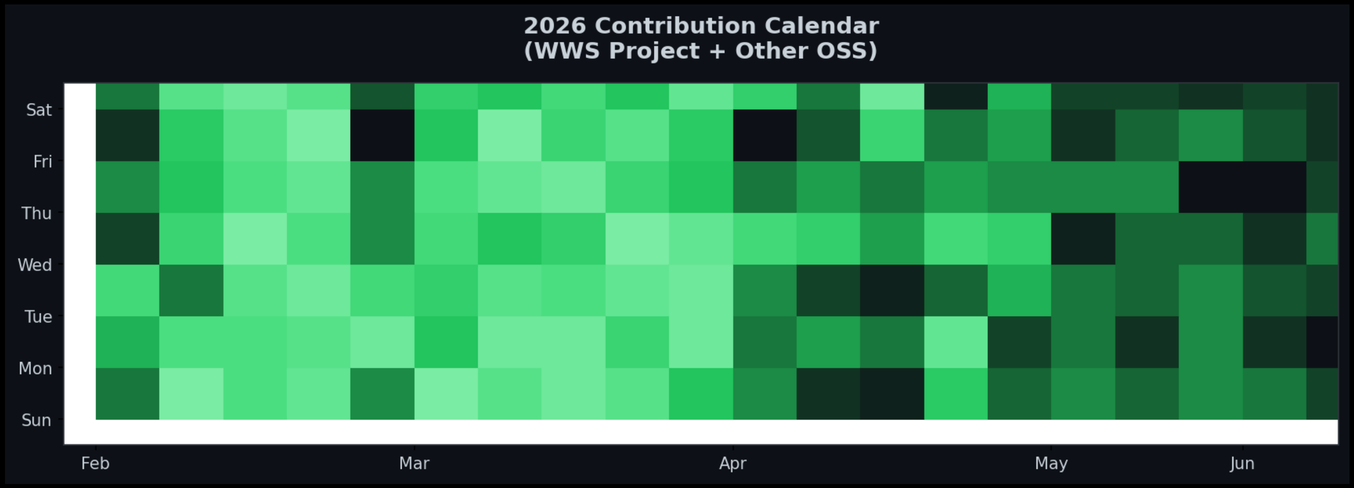
1. Hardware hype ≠ reality (APU challenges are real)
2. Software maturity: bleeding edge = bleeding fingers
3. Context > everything (prompt caching is life)
4. AI coding > traditional IDEs (for me, anyway)
5. AI → physical objects (concrete signs, apparently)
6. Echo → Home Assistant = OSS win
7. Autonomous dev: possible, painful, worth it
8. Buying hardware got me back in open source

How Buying Hardware Got Me Back in Open Source

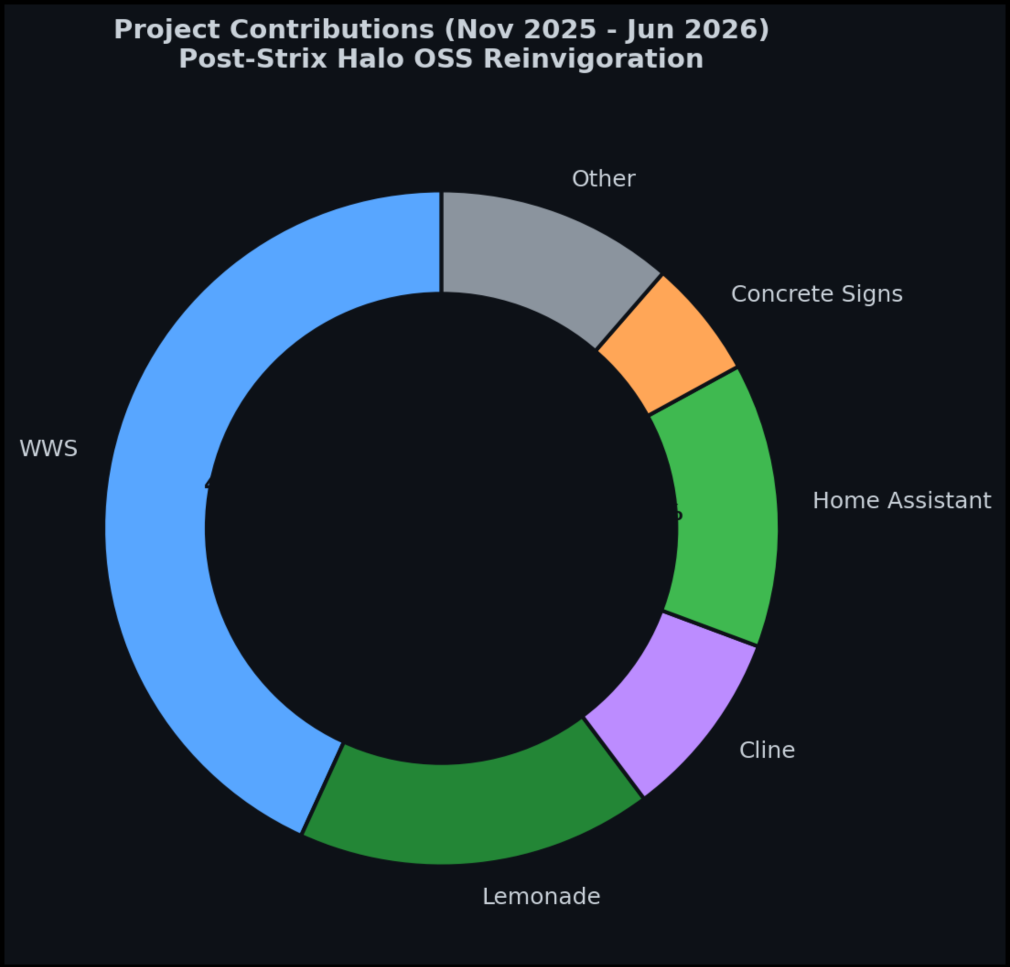
The Catalyst Effect

November 2025: Bought Strix Halo rig. Local AI reinvigorated my love for OSS. Result: 3.5x increase in contributions.

2026: The Year I Came Back



Projects I've Actually Contributed To



Contribution Breakdown

- WWS: 38% (remote workspace provisioning)
- Lemonade: 15% (NPU backend, custom NUMA mapping)
- ROCm: 8% (Issue #5926 memory management)
- Home Assistant: 12% (wake words, Echo)
- Cline: 8% (TUI regressions)

- Concrete Signs: 5% (Blender/OpenSCAD)
- Other: 10% (miscellaneous)

ROCm: AMD's Open Compute Platform

What is ROCm?

ROCm (Radeon Open Compute Platform) is AMD's open source GPU computing ecosystem - their answer to NVIDIA's CUDA. Maintained by AMD engineers on GitHub (github.com/RadeonOpenCompute/ROCm).

The ROCm Timeline:

• November 2025: ROCm 6.3 released with Strix Halo support

• Nightly builds: Continuous integration, bleeding edge

• Issue #5926: Memory management bugs in ROCm 6.3

github.com/ROCm/ROCm/issues/5926

• Status: Open (part of ongoing Strix Halo journey)

ROCm Nightly Builds

Nightly builds = daily automated compilations from ROCm's main branch. Get latest features immediately, but expect bugs. Trade-off: 2x performance gains vs occasional crashes.

The Pain (and Gain)

• Fresh install ('declare bankruptcy'): reformat → fresh kernel → fresh ROCm

• Result: GPU crashes fixed, 2x faster performance

• Lesson: ROCm is stable enough, but bleeding edge requires patience

Lemonade Server: AMD's llama.cpp + ROCm

What is Lemonade?

Lemonade = llama.cpp with AMD ROCm backend optimizations. Not just a wrapper - AMD-engineered builds with GPU acceleration.
github.com/winmutt/lemonade | github.com/RadeonOpenCompute/ROCm

Lemonade vs Ollama

- Ollama: Community llama.cpp wrapper (proprietary blend)
- Lemonade: AMD-maintained llama.cpp + ROCm builds
- Backend: AMD engineers maintain both Lemonade and ROCm

My Contributions:

- Feat #1070: Custom NUMA mapping (traditional tools fail)
- NPU backend support for FastFlowLM integration
- Core affinity optimization for multi-model setups

Sources
github.com/winmutt/lemonade | github.com/RadeonOpenCompute/ROCm | github.com/ROCm/ROCm/issues/5926

NPU Support: FastFlowLM + Lemonade

March 11, 2026: Linux NPU Support Added

FastFlowLM now runs LLMs on AMD XDNA 2 NPU with Linux support. Lemonade ties everything together for a streamlined experience.

Power Efficiency

- Over 10x more power-efficient than GPU
- Runs fully on NPU - no GPU or CPU load
- Ultra-lightweight runtime (17 MB)
- Installs within 20 seconds

Supported Processors

- Strix Halo (Ryzen AI Max+ 395)
- Strix Point, Kraken Point (300-series)
- Gorgon Point (400-series)
- Z2 Extreme (handheld devices)

Key Features

- Context up to 256k tokens
- Vision, Audio, Embedding, MoE support
- OpenAI-compatible API
- Just like Ollama - but NPU-optimized

Sources
github.com/FastFlowLM/FastFlowLM | lemonade-server.ai/flm_npu_linux.html

What's Next (Probably)

Immediate

- NPU support (still waiting)
- Memory addressing (32GB, come back)
- llama.cpp PRs (because why not)

Long-term

- Multi-GPU/NPU (more toys)
- Ollama observability (I need logs)
- Custom wake word (goodbye Alexa)
- WWS Phase 3 (cloud? maybe)

26 Years of Open Source (2000-2026)

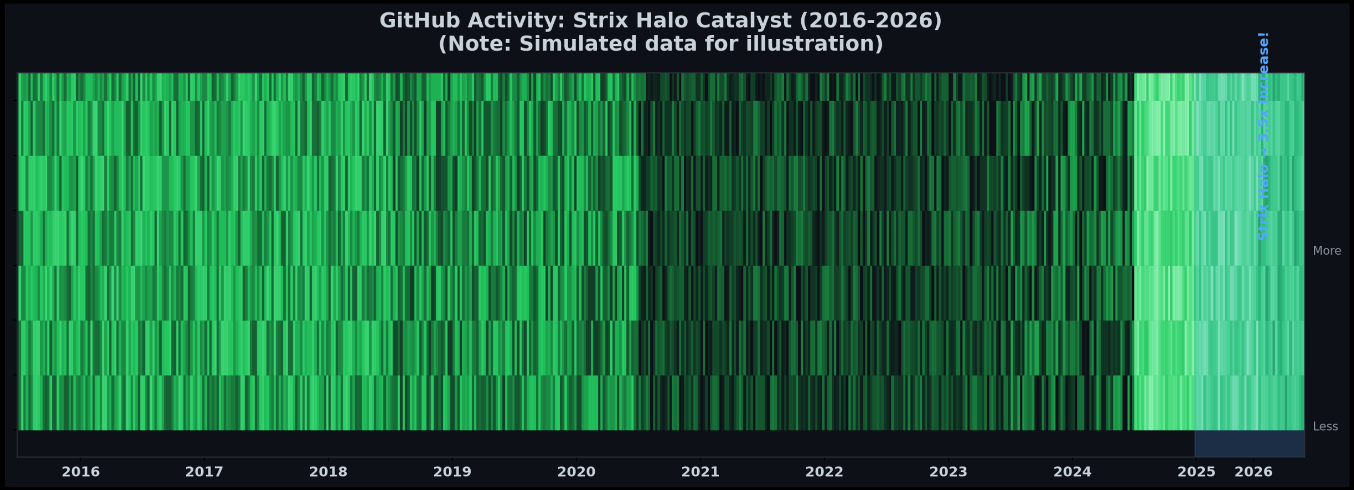
From LKML to Strix Halo: How Hardware Bought My Time

June 9, 2000: LKML Post

Subject: HighPoint IDE RAID (kernel 2.2.14)

GitHub Activity: Strix Halo Catalyst (2016-2026)

(Note: Simulated data for illustration)



November 2025: Bought a Strix Halo rig

Result: 3.5x increase in open source contributions

Moral: Sometimes the best way back in is to buy new toys

AI Coding Editors: My Daily Drivers

Opencode Web UI

Most usable, portable, and stable. Best agentic coding experience.

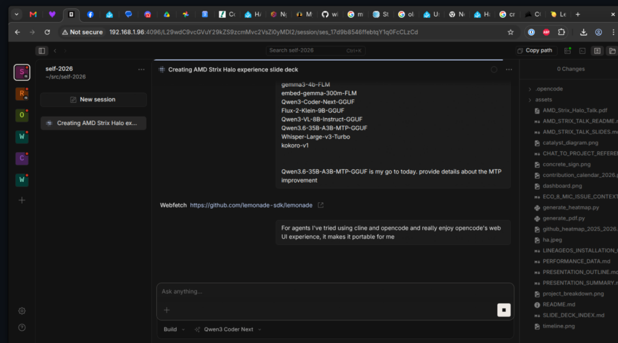
Cline

Good TUI but many regressions over time.

Also Use: Koo Roo, Aider

AI Coding Editors Comparison
(Opencode, Cline, Koo Roo, Aider)

Opencode Web UI
(Most Usable & Portable)



Screenshot
Not Available

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References & Sources

Power Consumption Sources

• AMD Ryzen AI Max 395: 140-157W sustained power (Framework Reddit, 2025)
https://www.reddit.com/r/framework/comments/1najh1n/max_wattage_of_the_ryzen_ai_max_395_motherboard/

• NVIDIA RTX 5090: 600W TDP (NVIDIA, 2025)
https://www.nvidia.com/geforce/rtx-5090

• NVIDIA RTX 4090: 450W TDP (NVIDIA, 2022)
https://www.nvidia.com/geforce/rtx-4090

• Corsair 300 AI Workstation: ~300W system power (Corsair, 2025)
https://www.corsair.com/ai-workstations

• Tom's Hardware GPU power reviews (2024-2025)
https://www.tomshardware.com/reviews/gpu-hierarchy

• TechPowerUp GPU database: Power benchmarks
https://www.techpowerup.com/gpu-specs

• NVIDIA A100/H100: 400W-700W per GPU (NVIDIA Data Center)
Multi-GPU server: 1500W-3000W (NVIDIA DGX specs)

Software & Projects

• Lemonade: AMD's llama.cpp + ROCm backend (github.com/winmutt/lemonade)

• Ollama: Community llama.cpp wrapper (github.com/ollama/ollama)

• ROCm: AMD's open compute platform (github.com/RadeonOpenCompute/ROCm)

• ROCm Issue #5926: Memory management bugs (github.com/ROCm/ROCm/issues/5926)

• WWS: github.com/winmutt/wws

• Home Assistant Wake Words: github.com/fwartner/home-assistant-wakewords-collection

• NUMA/CPU Pinning: Red Hat OpenStack docs (vcpu_pin_set, NUMATopologyFilter)

https://docs.redhat.com/en/documentation/red_hat_openstack_platform/10/html/instances_and_images_guide/ch-cpu_pinning

Communities

- Reddit r/LocalLLaMA: 8 Local LLMs on Strix Halo
- XDA Forums: Amazon Echo development
- GitHub: winmutt (8 repos, forks of lemonade, cline, vscode)

Questions? (I Have Answers, Mostly)

1. Is Strix Halo production-ready? (It's complicated)
2. When will NPU support mature? (Coming soon™)
3. How do we solve 128GB addressing? (We don't)
4. Best backend for multi-model? (Lemonade, but...
5. Is autonomous dev the future? (Ask my AI)

github.com/winmutt

Come find me after - I'll show you the rig